

EX. NO: 6 — IMPLEMENTATION OF DES

Aim:

To implement the Data Encryption Standard (DES) encryption technique using C.

Algorithm:

1. Read the plaintext.
2. Read the key.
3. Perform XOR operation between plaintext and key.
4. Display the encrypted text.
5. Perform XOR again to decrypt.
6. Display the decrypted text.

Program:

```
#include <stdio.h>

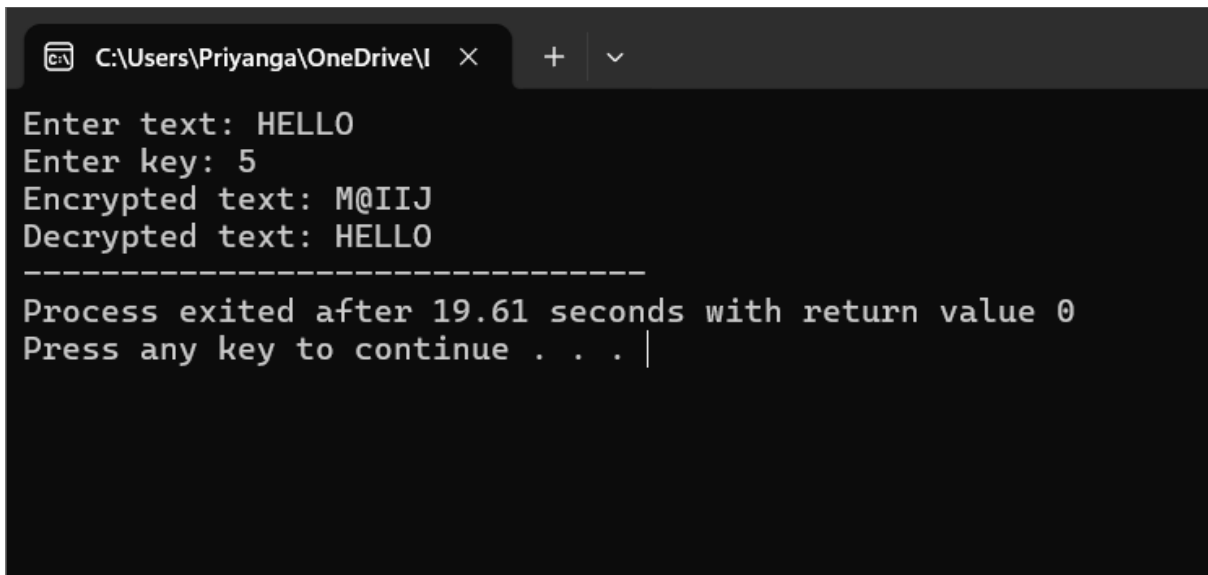
int main()
{
    char text[100], encrypted[100], decrypted[100];
    int key, i;
    printf("Enter text: ");
    scanf("%s", text);
    printf("Enter key: ");
    scanf("%d", &key);
    /* Encryption */
    for(i = 0; text[i] != '\0'; i++)
    {
        encrypted[i] = text[i] ^ key;
    }
    encrypted[i] = '\0';
    printf("Encrypted text: ");
    for(i = 0; encrypted[i] != '\0'; i++)
```

```

        printf("%c", encrypted[i]);
    /* Decryption */
    for(i = 0; encrypted[i] != '\0'; i++)
    {
        decrypted[i] = encrypted[i] ^ key;
    }
    decrypted[i] = '\0';
    printf("\nDecrypted text: %s", decrypted);
    return 0;
}

```

Output:



```

C:\Users\Priyanga\OneDrive\I >
Enter text: HELLO
Enter key: 5
Encrypted text: M@IIJ
Decrypted text: HELLO
-----
Process exited after 19.61 seconds with return value 0
Press any key to continue . . . |

```

Result:

Thus, the encryption and decryption technique was implemented successfully using C.